Supplementary Material for:

The biosecurity collective-risk social dilemma: Simulating the prevention of a rapidly spreading pest

Abstract

This supplementary document reports additional details of the study by *authors masked for peer review* examining the effect of risk and communication on biosecurity decisions in the biosecurity collective-risk social dilemma. It includes the virtual lottery game play panels, experimental instructions and control questions, as well as supporting game-theoretic analyses, statistical results, tables, and figures. This document is not a stand-alone paper and hence is not intended to be self-explanatory—please refer to *authors masked for peer review* for further explanation.

Supplementary Virtual Lottery Game

Prize structure for Play Panel 1 (Games 1–14).

Game	Ball 1–3	Ball 4–10	Choice	Ball 1	Balls 2–10	Choice
	Prize A			Prize B		
1	$40	$10	○	$68	$5	○
2	$40	$10	○	$75	$5	○
3	$40	$10	○	$83	$5	○
4	$40	$10	○	$93	$5	○
5	$40	$10	○	$106	$5	○
6	$40	$10	○	$125	$5	○
7	$40	$10	○	$150	$5	○
8	$40	$10	○	$185	$5	○
9	$40	$10	○	$220	$5	○
10	$40	$10	○	$300	$5	○
11	$40	$10	○	$400	$5	○
12	$40	$10	○	$600	$5	○
13	$40	$10	○	$1000	$5	○
14	$40	$10	○	$1700	$5	○

Prize structure for Play Panel 2 (Games 15–28).

Game	**Ball 1–9**	**Ball 10**	**Choice**	**Balls 1–7**	**Balls 8–10**	**Choice**
	Prize A			**Prize B**		
15	$40	$30	○	$54	$5	○
16	$40	$30	○	$56	$5	○
17	$40	$30	○	$58	$5	○
18	$40	$30	○	$60	$5	○
19	$40	$30	○	$62	$5	○
20	$40	$30	○	$65	$5	○
21	$40	$30	○	$68	$5	○
22	$40	$30	○	$72	$5	○
23	$40	$30	○	$77	$5	○
24	$40	$30	○	$83	$5	○
25	$40	$30	○	$90	$5	○
26	$40	$30	○	$100	$5	○
27	$40	$30	○	$110	$5	○
28	$40	$30	○	$130	$5	○

Prize structure for Play Panel 3 (Games 29–35).

Game	Balls 1–5	Balls 6–10	Choice	Balls 1–5	Balls 6–10	Choice
	Prize A			Prize B		
29	$25	-$4	○	$30	-$21	○
30	$4	-$4	○	$30	-$21	○
31	$1	-$4	○	$30	-$21	○
32	$1	-$4	○	$30	-$16	○
33	$1	-$8	○	$30	-$16	○
34	$1	-$8	○	$30	-$14	○
35	$1	-$8	○	$30	-$11	○

Supplementary Experimental Instructions

No-Communication Treatment

Welcome to the Experiment!

In this experiment, you can earn money. **For a successful experiment, you must not communicate with other participants or make yourself noticeable to them in any way**. Please read the following instructions carefully. Should you have any questions, please signal us. Thank you for your participation. We hope you enjoy the experiment, and wish you good luck!

After reading please click 

Today's session

Today's session is divided into three parts: (1) a virtual lottery game, (2) a biosecurity game, and (3) a questionnaire. We will begin with the virtual lottery game.

After reading please click  to proceed to the instructions for the virtual lottery game.

Part 1: Virtual Lottery Game

Instructions

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The PLAY PANELS contain the odds of each Prize, for each game. Please close the instructions momentarily by clicking on the blue instructions icon to the right of the screen and you will be able to see the first of three PLAY PANELS. Once you have inspected the PLAY PANEL, click on the instructions icon again to return back to these instructions.

In each PLAY PANEL, the default choice is Prize A for each game. If you find the odds in Prize B more appealing at any given game, you can switch to Prize B. However, once you've switched to Prize B, you cannot go back to Prize A for the rest of the PLAY PANEL.

How the game works

Thirty-five balls (numbered 1-35) will be put into a virtual bingo cage and one ball will be randomly picked. This numbered ball will correspond to the GAME NUMBER in one of your PLAY PANELS and determines which out of the 35 games will be played.

All balls are then removed from the bingo cage and ten balls (numbered 1-10) are put back in. The next ball that is picked determines how much money you win (or lose).

For example, if GAME NUMBER 7 was randomly selected out of the 35 games, and you chose Prize A for that game, you will win:

- $40 if the ball number is between 1 and 3, or
- $10 if the ball number is between 4 and 10.

If you chose Prize B, you will win:

- $150 if the ball number is 1, or
- $5 if the ball number is between 2 and 10.

When instructed to do so, please click  to proceed

Part 2: Biosecurity Game

Overview of the game

Please read through these instructions carefully. If you have any questions please raise your hand and one of the administrators will come to you.

You are about to take part in an experiment about collective decision-making related to risky choices. This experiment gives you an opportunity to earn money. How much you earn depends partly on your decisions, partly on the decisions of others, and partly on chance.

In this experiment, you own an apple orchard and you have three neighbours who also grow apples. Each season, your orchard will generate revenue. You will receive this revenue in the seasons where there is ***no outbreak of pests***. When an ***outbreak*** of pests occurs, the pests spread quickly and infect all neighbouring orchards, resulting in the loss of that season's apples. You will generate no revenue in a season where an ***outbreak*** of pests occurs.

You must decide whether to ***protect*** or ***not protect*** against an ***outbreak*** of pests (at the start of) each season. There is a cost associated with choosing to ***protect***. You do not have to pay this cost if you choose to ***not protect***.

The probability of an ***outbreak*** of pests occurring is quite variable and partially dependent on whether your neighbours protect their orchards. Sometimes, if everyone chooses to ***not protect*** there will be ***no outbreak*** of pests. On other occasions, the majority may choose to ***protect***, and an ***outbreak*** of pests will still occur.

After reading please click 

Your anonymity

All decisions you make in the game are anonymous. To ensure this, a pseudonym is assigned to you by the computer, visible at the bottom left of the screen. These pseudonyms correspond to names of moons in our solar system (Leda, Triton, Portia, or Sinope).

Game structure

The game is divided into two blocks of 15 rounds (30 rounds in total), where each round corresponds to a season. Each round is divided into two stages: a contribution stage and a feedback stage.

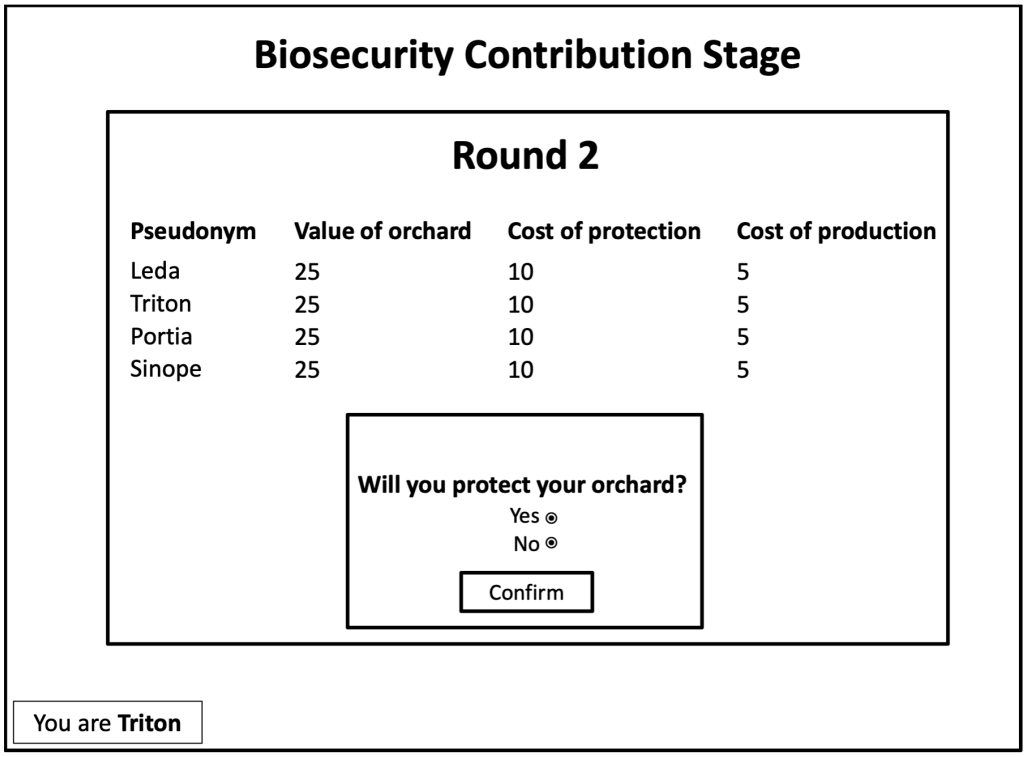


Figure 1. Illustration of the contribution stage of the biosecurity game.

A screenshot of the contribution stage is provided in Figure 1. In this stage, you will be presented with three sources of information about each player for the current playing round: the value of their orchard, their cost of protecting it, and their cost of production.

After reading please click 

The value of each player's orchard represents the amount of revenue it will generate if there is no outbreak of pests. The cost of protection represents each player's cost of protecting their orchard from an outbreak of pests. The cost of production represents each player's cost of producing his or her apples. For simplicity, in today's game, the value of each player's orchard ($25), their cost of protection ($10), and their cost of production ($5) are identical and do not vary across rounds.

In the contribution stage, you must decide whether you will protect your orchard from an outbreak of pests or not. If you select "Yes" then you will be required to pay the cost of protection of $10, whereas if you select "No" then you will not be required to pay this cost.

Note that whilst you can choose on each round whether or not to protect your orchard, you will always be required to pay the $5 cost of production. This will be automatically deducted from any revenue you earn on each round.

Your revenue on each round is determined as follows:

- If you choose to ***protect*** and there is ***no outbreak***, the income you receive is **$10** (value of orchard - cost of protection - cost of production = 25 – 10 – 5 = 10).
- If you choose to ***not protect*** and there is ***no outbreak***, the income you receive is **$20** (value of orchard - cost of production = 25 – 5 = 20).
- If you choose to ***protect*** and there is an ***outbreak***, the loss you receive is **-$15** (0 - cost of protection - cost of production = 0 – 10 – 5 = -15).
- If you choose to ***not protect*** and there is an ***outbreak***, the income you receive is **-$5** (0 – cost of production = 0 – 5 = -5).

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After all players have simultaneously made their decision whether or not to protect their orchards in the contribution stage, the computer will proceed to the feedback stage. A screen shot of this stage is provided in Figure 2.

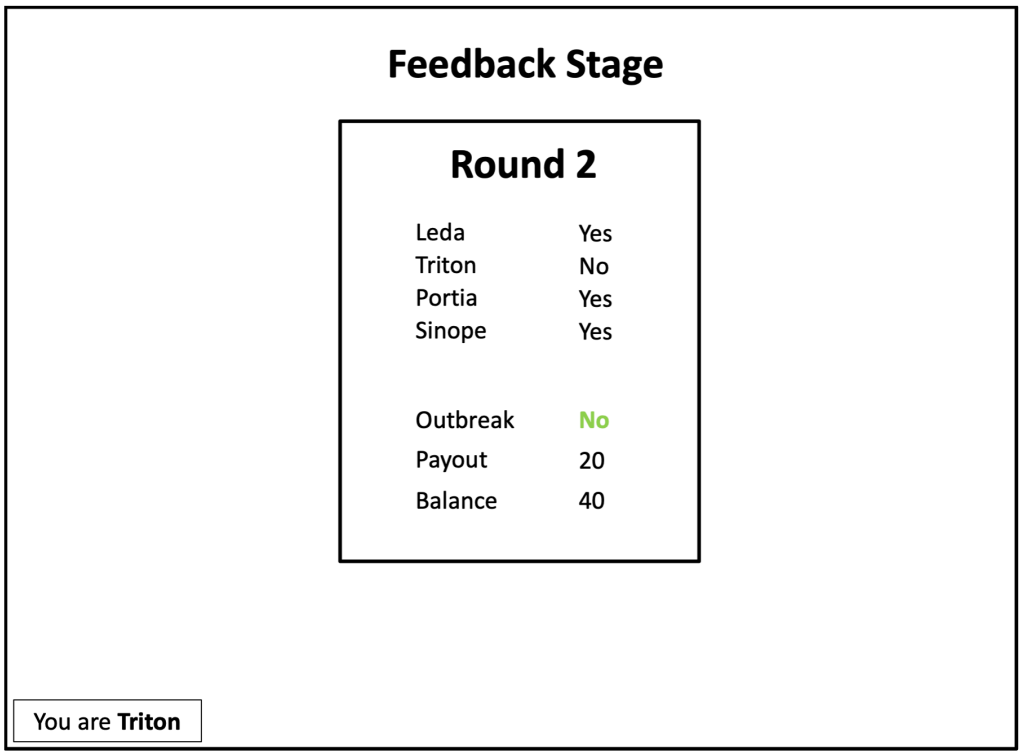


Figure 2. Illustration of the feedback stage of the biosecurity game.

In the feedback stage, you will receive information about the decisions made by each player in the contribution stage. In the example shown in Figure 2, it can be seen that on Round 2 three players decided to incur the cost of protecting their orchards (Leda, Portia, and Sinope) and one player decided not to incur that cost (Triton). You will also receive information about whether an outbreak of pests occurred, your payout for the round, and the balance of your account so far. In the example, an outbreak did not occur and the player (Triton) earned revenue of $20, taking the balance of her account to $40, meaning that she must also have earned revenue of $20 on Round 1.

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Probability of an outbreak of pests

As noted earlier, you will complete two blocks of 15 rounds of the biosecurity game. In one block, the probability of an outbreak of pests will be low (low-risk condition), whereas in the other block the probability of an outbreak of pests will be high (high-risk condition).

Low-risk condition. The table below gives the probability of an outbreak on each round in the low-risk condition, as a function of the number of players choosing to protect, or not protect, their orchards.

Number of players choosing to:		Probability of an ***Outbreak %***
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Your payout

At the end of the experiment, you will be paid any revenue generated during the biosecurity game, subject to a conversion rate. The conversion rate is your revenue earned $\times$ 0.1. For example, if your revenue earned was \$300 then you would be paid 300 $\times$ 0.1 = \$30.

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Player Strategies

You can choose to play using a number of different strategies.

For example, one strategy would be for each player to choose to protect on every round in both the low- and high-risk blocks. Using this strategy, each player would earn a **total of $30:** $15 in the low-risk block and $15 in the high-risk block.

Another strategy would be for all players to choose to not protect on each round. Using this strategy, each player's earnings would depend upon the number of pest outbreaks that occurred.

Of course, there are many other strategies you could use aside from these two examples.

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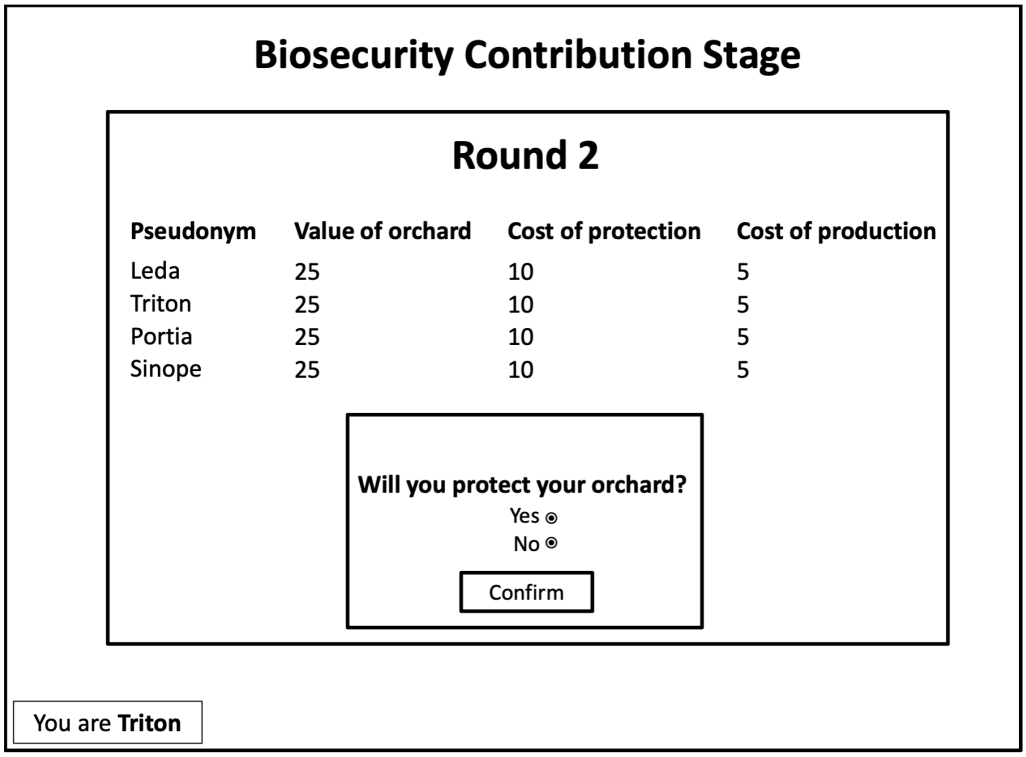


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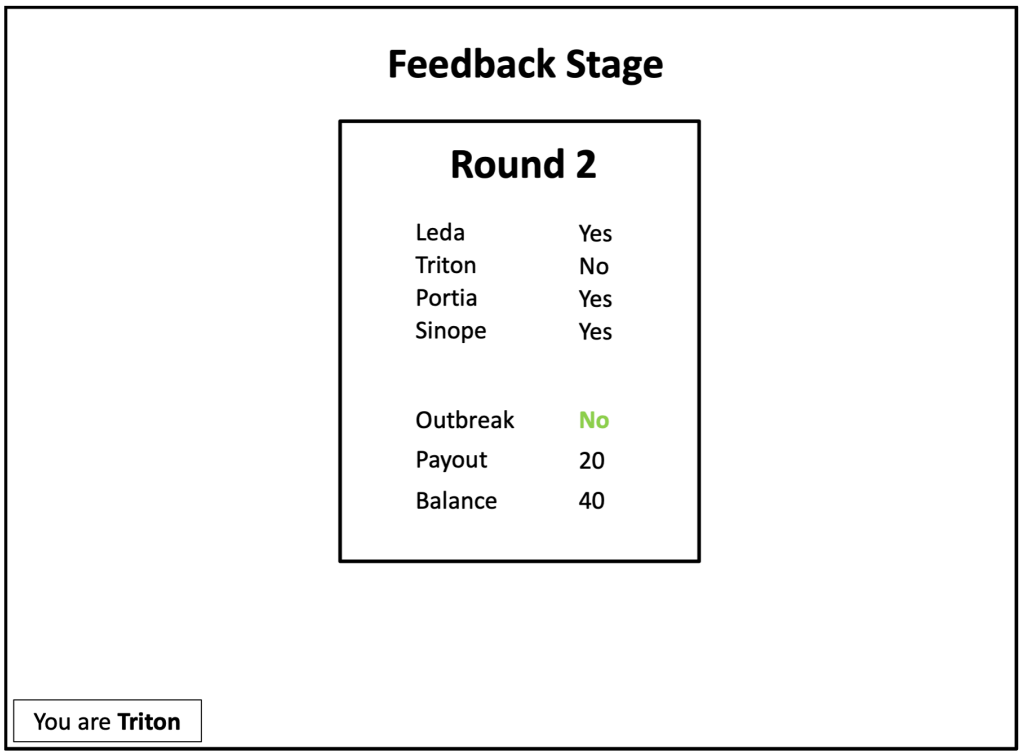


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After reading please click →

Communication Stage

At the beginning of **rounds 1, 6,** and **11** in the low- and high-risk blocks, you will be able to communicate with your co-players via a "chat room" before you enter the biosecurity contribution stage. A screenshot of this chat room is provided in Figure 3.

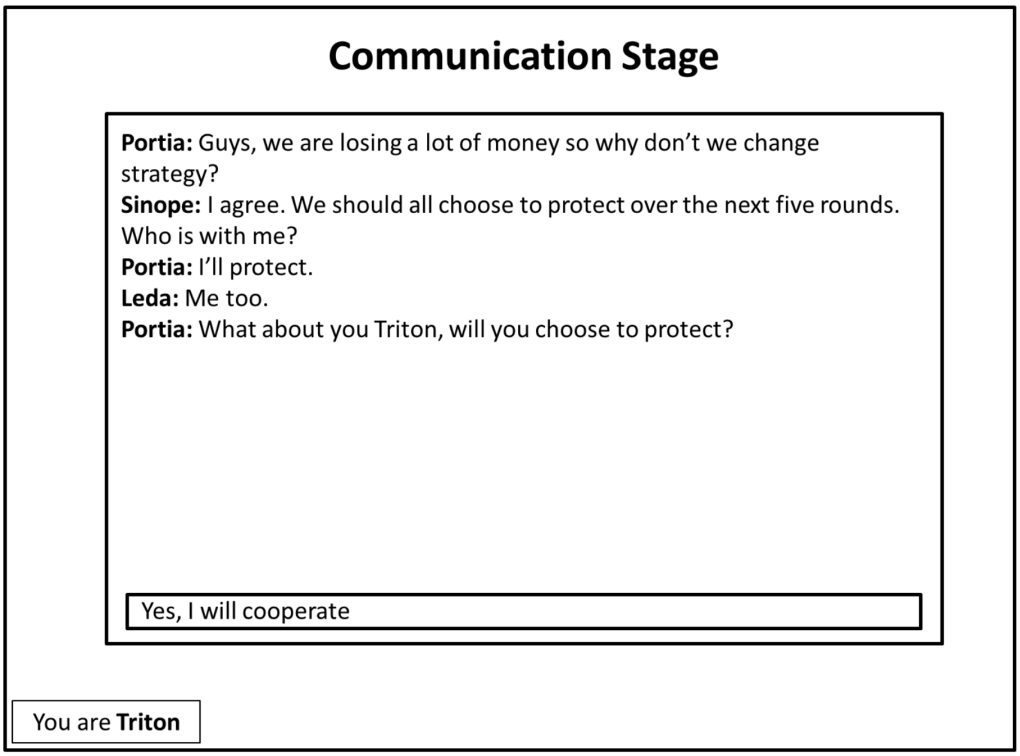


Figure 3. Illustration of the communication stage of the biosecurity game.

The chat room contains a text entry field at the bottom (in the example in Figure 3, this is the field in which the player Triton has typed—but not yet submitted—the message "Yes, I will cooperate"). You can type a message in this field and when you press the enter key on the keyboard it will be displayed in the public window under your assigned pseudonym for the other players to see. In the chat room, you can discuss anything, but please abide by the following rules:

- Refrain from using obscene language and do not threaten other players
- Do not offer side payments. An example of a side payment would be if Portia said:

"Triton, if you choose to protect over the next five rounds, I'll give you $5 of my experimental earnings"

- Keep your real identity private at all times

After reading please click 

Remember that you will only be able to communicate with your co-players through the chat room at the beginning of **rounds 1, 6,** and **11**. On each of these rounds, you will have **exactly 2 minutes** to communicate with one another in the chat room.

After reading please click 

Probability of an outbreak of pests

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Supplementary Control Questions

Player Pseudonym: _____ **Condition:** _____ **Group #:** _____

Control Questions

For a successful experiment, it is imperative that each player understands the rules of play. Please answer the following questions so that we can ensure that you have understood the instructions. Please answer all questions, and signal us when you have finished. We will then come to you and check your answers.

1. How many players must protect their orchards in order to prevent an outbreak of pests with certainty (please tick the correct box)?

   ☐ 1 player

   ☐ 2 players

   ☐ 3 players

   ☐ 4 players

2. Assume that you are in the high risk condition. If two players decide to protect and another two players decide not to protect, then what is the probability of an outbreak (please tick the correct box)?

   ☐ 0%

   ☐ 20%

   ☐ 36%

   ☐ 59%

3. Assume that you are in the low risk condition. If one player decides to protect and three players decide not to protect, then what is the probability of an outbreak (please tick the correct box)?

☐ 0%

☐ 19%

☐ 27%

☐ 34%

4. Assume that you decided to protect your orchard and there was an outbreak of pests. What would your revenue be (please tick the correct box)?

☐ -$15

☐ -$5

☐ $10

☐ $20

5. Assume that you decided not to protect your orchard and there was no outbreak of pests. What would your revenue be (please tick the correct box)?

☐ -$15

☐ -$5

☐ $10

☐ $20

When you have finished answering the questions, please signal us so that we can come to you and check your answers.

Supplementary Game-Theoretic Analysis

Extending Hennessy's One-Shot Game to a Repeated Game

Applying Hennessy's one-shot game-theoretic model to the richer experimental biosecurity game analysed in this paper requires the model to be extended to account for learning during a repeated game with incomplete information about the preferences and strategies of other players. Fudenberg and Tirole (1991, p.331) classify this game as a repeated game with observed actions and incomplete information. It has been analysed by a number of authors as a finitely repeated prisoner's dilemma (Andreoni & Miller, 1993; Axelrod, 1981; Kreps et al., 1982).

In the biosecurity game, the possible actions are independent of a player's type and are such that $b_i \in \{0, 1\}$; that is, a player either biosecures or not. This decision set is available to all players in all rounds. The vector of all actions in period t is $b^t = (b_1^t, \ldots, b_N^t)$ and $h^t = (b^1, \ldots, b^{t-1})$ gives the history of actions at the beginning of round t.

A repeated game allows for learning that is reinforced in two of the treatments by the communication (cheap talk) intervention. Even without communication, players are fully informed about the contributions of other players in the group in the previous periods. Therefore, players signal their type by choosing to biosecure or not. This may include their attitude towards cooperation and risk preferences.

For instance, as we establish below for a two-player game, a rational player with a high degree of loss aversion will frame the game as a coordination game rather than a prisoner's dilemma. They will signal their type by cooperating in most rounds and appear to be an unconditional cooperator. A type of player who has adopted an inflexible tit-for-tat strategy will biosecure (cooperate) until they observe the other player not biosecure and resume when they see the other player biosecure. In this setting, a tit-for-tat strategy is similar to conditional cooperation.

Kreps et al. (1982) establish an important result that if a rational player assigns a

probability that the other player will play tit-for-tat (conditional cooperation), then this is sufficient to ensure cooperation in a prisoner's dilemma in most rounds of the game, apart from the last one or two. If both players are strictly rational, then by backward induction the Nash equilibrium for the game is to *Not Biosecure* in any round. This equilibrium does not explain the relatively high level of cooperation observed in most experimental studies of repeated prisoner's dilemma games (Andreoni & Miller, 1993; Axelrod, 1981; Kreps et al., 1982). Our results also show significant cooperation in the high-risk prisoner's dilemma game and the low-risk no dilemma game, where Not Biosecure is an optimal cooperative solution as well as a Nash equilibrium (see Figure 1). Cooperation may also be explained by risk preferences and signalling strategies.

Strategic Types

In this sub-section we assume risk neutrality and that loss aversion $\lambda = 1$. Suppose the two players COLUMN and ROW could follow a tit-for-tat strategy (conditional cooperation) with some probability (Table S1). Selecting Not Biosecure will immediately signal that they are playing a different strategy. The payoffs are such that the game is a repeated prisoner's dilemma, thus $BB > NN$, $NB > BN$, $NB > BB$, $NB > NN$ and $NB + BN < 2BB$ (where B = Biosecure; N = Not Biosecure). The last condition is to prevent cooperation from being Pareto-dominated by an alternating strategy where players take turns to Biosecure.

For instance, if both players play tit-for-tat, Biosecure will be observed in all remaining periods T, including the last. The payoff for each of these players would therefore be BB. The most that a "rational" player could gain by playing Not Biosecure against a tit-for-tat player would be $NB - NN + T(NN - BB)$. By assumption, $NB - NN > 0$, but $T(NN - BB) < 0$. Therefore, the number of rounds remaining is critical to the decision to Biosecure.

Player Risk Attitudes

The biosecurity game involves gains when there is no outbreak and losses when there is an outbreak. If individual behaviour is represented by cumulative prospect theory, this would imply a two-part utility function $u_i(\pi)$; for instance, the power function proposed by Tversky and Kahneman (1992):

$$u_i(\pi) = \begin{cases} \pi^{\alpha_i} & \text{if } \pi \geq 0 \\ -\lambda_i(-\pi)^{\beta_i} & \text{if } \pi < 0 \end{cases} \tag{1}$$

where π gives the gain or loss in a period. For the biosecurity game, the expected utility from the decision $b_i^t \in \{0, 1\}$, given that $b_{-1} < N - 1$ other players contribute, is:

$$E[u(b_i^t|b_{-i}^t)] = \sigma^{N-b_{-i}^t-b_i^t} u_i(v_i - c_i^p - b_i^t c_i^b) + (1 - \sigma^{N-b_{-i}^t-b_i^t}) u_i(-c_i^p - b_i^t c_i^b)$$

The first term is assumed to be positive, while the second is non-positive. If the player knows b_{-i}^t with certainty, they solve the following optimisation problem for a one-shot biosecurity game:

$$b_i^t \in \underset{b_i^t \in \{0,1\}}{\operatorname{argmax}} \, \sigma^{N-b_{-i}^t-b_i^t} u_i(v_i - c_i^p - b_i^t c_i^b) + (1 - \sigma^{N-b_{-i}^t-b_i^t}) u_i(-c_i^p - b_i^t c_i^b)$$

For the biosecurity game, player types can be defined in terms of their attitudes towards cooperation (conditional cooperator or rational) and in terms of their risk and loss aversion.

Applying this to the utility function (1) (dropping the time subscript for clarity):

$$V(\sigma, b_{-i}, b_i, \alpha_i, \lambda_i) = \sigma^{N-b_{-i}-b_i}(v_i - c_i^p - b_i c_i^b)^{\alpha_i} - \lambda_i(1 - \sigma^{N-b_{-i}-b_i})(c_i^p + b_i c_i^b)^{\alpha_i}$$

Based on this function, the piecewise certainty equivalent function can be given by:

$$CE(\sigma, b_{-i}, b_i, \alpha_i, \lambda_i) = \begin{cases} (V(\sigma, b_{-i}, b_i, \alpha_i, \lambda_i))^{\frac{1}{\alpha_i}} & \text{if } V(\sigma, \alpha_i) \geq 0 \\ -\left(\frac{-V(\sigma, b_{-i}, b_i, \alpha_i, \lambda_i)}{\lambda_i}\right)^{\frac{1}{\alpha_i}} & \text{if } V(\sigma, \alpha_i) < 0 \end{cases} \quad (2)$$

The incentive to biosecure can be measured in terms of differences in the certainty equivalents:

$$\Delta_{CE} = CE(\sigma, b_{-i}, 1, \alpha_i, \lambda_i) - CE(\sigma, b_{-i}, 0, \alpha_i, \lambda_i) \quad (3)$$

General results on the incentives are difficult to derive. However, if $(N - b_{-i} - b_i) = 0$, taking the derivatives of equation (3) it can be shown that the incentive to biosecure is strictly increasing in the level of constant relative risk aversion, $1 - \alpha_i$, and in loss aversion if the certainty equivalent of not biosecuring is positive.

Specific Example

Attitude to risk is fundamental as it can change the structure of the game to the extent that a rational, loss averse player may perceive a coordination game rather than a prisoner's dilemma. For instance, suppose there are just two types: a risk-neutral (RN) type, for whom $\alpha_i = \beta_i = \lambda_i = 1$, and a loss-averse (LA) type with the same set of parameters except that $\lambda_i > 1$. Consider a two-player, two-type example based on the parameters from the two-player biosecurity game. If both players are of the loss-averse type, with $\lambda_i = 2.7$ (the loss aversion parameter value proposed by Tversky and Kahneman, 1992), the game becomes a coordination game with the payoffs given in Table S2.

Taking the loss averse utility function and applying it to equation (3) in the paper, the condition for a loss-averse individual biosecuring is:

$$\sigma^{N-b_{-i}-b_i}(v_i - c_i^p - c_i^b) - \lambda(1 - \sigma^{N-b_{-i}-b_i})(c_i^p + c_i^b) \geq \sigma^{N-b_{-i}}(v_i - c_i^p) - \lambda(1 - \sigma^{N-b_{-i}})(c_i^p)$$

Case 1, if $N = 2$, $b_{-i} = 1$, and we write $\lambda = 1 + \epsilon$, where $\epsilon = \lambda - 1$ denotes the degree of loss aversion above the baseline of $\lambda = 1$, then by biosecuring the player can eliminate the risk and guarantee the payoff $(v_i - c_i^p - c_i^b)$:

$$(v_i - c_i^p - c_i^b) \geq \sigma(v_i - c_i^p) - (1 + \epsilon)(1 - \sigma)(c_i^p)$$

Simplifying to apply Hennessy's criteria for biosecurity gives:

$$(1 - \sigma) \geq (c_i^b - \epsilon c_i^p(1 - \sigma))/v_i \leq c_i^b/v_i$$

The first result is that if $\epsilon > 0$, that is, the individual is loss averse, the incentive to biosecure is increased relative to the condition under risk neutrality, where $\epsilon = 0$.

Interestingly, this result is reversed if $N = 2$ and $b_{-1} = 0$:

$$\sigma(v_i - c_i^p - c_i^b) - (1 + \epsilon)(1 - \sigma)(c_i^p + c_i^b) \geq \sigma^N(v_i - c_i^p) - (1 + \epsilon)(1 - \sigma^N)(c_i^p)$$

The incentive to biosecure with loss aversion is weakened relative to risk neutrality:

$$\sigma(1 - \sigma) \geq (c_i^b + \epsilon(1 - \sigma)(c_i^b - c_i^p\sigma))/v_i \gtreqless c_i^b/v_i$$

If $\epsilon(1 - \sigma)(c_i^b - c_i^p\sigma) > 0$, the sign of this term hinges on the relative risk associated with the biosecurity cost and the production cost, $(c_i^b - c_i^p\sigma)$. For instance, for the high-risk game parameters, this term is $(10 - 0.8 \times 5) > 0$. Thus, a loss-averse player has a smaller incentive to biosecure, and this incentive declines with the level of loss aversion represented by ϵ.

Concluding Remarks

In an experimental game in which uncertainty is central, risk attitudes will play a role. In summary, higher levels of risk and loss aversion will tend to reinforce the decision to biosecure. However, this result depends on how much cooperation there is in the group. For instance, a loss-averse individual may be less inclined to biosecure if they perceive that their group will not contribute. This is because biosecuring increases the size of the loss in the case of an outbreak.

These results call into question the simple definition of types in public good games (Fischbacher et al., 2001; Ostrom, 2000), as conditional cooperation may be observed either due to an innate preference to cooperate or a particular set of preferences that change the structure of the game. This complication does not arise in a deterministic public goods game.

Supplementary Statistical Analyses

Selection of Random-Effects Structure

We selected the random-effects structure for the treatment-only model predicting biosecurity decisions using a two-stage model comparison procedure. All models included fixed effects for risk, communication, and round, together with all two-way and three-way interactions among these predictors. Because risk and round varied within groups and players, whereas communication varied between groups and players, we considered random slopes for risk, round, and their interaction, but not for communication.

We first compared alternative player-level random-effects structures while retaining random intercepts for both group and player. Specifically, we compared models with random intercepts only, a by-player random slope for risk, a by-player random slope for round, by-player random slopes for risk and round, and by-player random slopes for risk, round, and their interaction. As shown in Table S3, models with by-player random slopes provided a substantially better fit than the random-intercepts-only model. The best-fitting player-level structure included by-player random slopes for risk and round, but not their interaction. This model had the lowest AIC and BIC among the player-level models considered.

Holding this player-level structure constant, we then compared alternative group-level random-effects structures. Specifically, we compared models with a random intercept for group only, a by-group random slope for risk, a by-group random slope for round, by-group random slopes for risk and round, and by-group random slopes for risk, round, and their interaction. As shown in Table S4, adding group-level random slopes further improved model fit. AIC favoured the most complex group-level structure, with by-group random slopes for risk, round, and their interaction, whereas BIC favoured the more parsimonious model including a by-group random slope for round only. Given the relatively small number of groups and the stronger penalty for complexity under BIC, we

retained the model with a by-group random slope for round and by-player random slopes for risk and round as the final random-effects structure. The resulting structure was therefore:

$$(1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player})$$

This random effects structure was used for the treatment-only model, as well as the additive-feedback, social-feedback, and outcome-feedback models predicting biosecurity decisions.

Social-Feedback Model Selection

The social-feedback model inherited the structure of the additive-feedback model but allowed the effect of lagged protection to vary as a function of risk and communication, permitting an assessment of whether responsiveness to others' prior protection decisions varies by treatment. To determine the appropriate model specification, we fit two models. The first model included the two-way interactions of lagged protection with risk and communication. Formally, the two-way interaction model specification was:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{round} \\ & + \text{risk} : \text{communication} + \text{risk} : \text{round} + \text{communication} : \text{round} \\ & + \text{risk} : \text{communication} : \text{round} \\ & + \text{lagged_outcome} + \text{lagged_protection} \\ & + \text{risk} : \text{lagged_protection} \\ & + \text{communication} : \text{lagged_protection} \\ & + (1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player}) \end{aligned} \tag{3}$$

The second model included the same terms as the two-way interaction model, with

the addition of the three-way interaction between risk, communication, and lagged protection. Formally, the three-way interaction model specification was:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{round} \\ & + \text{risk} : \text{communication} + \text{risk} : \text{round} + \text{communication} : \text{round} \\ & + \text{risk} : \text{communication} : \text{round} \\ & + \text{lagged_outcome} + \text{lagged_protection} \\ & + \text{risk} : \text{lagged_protection} \\ & + \text{communication} : \text{lagged_protection} \\ & + \text{risk} : \text{communication} : \text{lagged_protection} \\ & + (1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player}) \end{aligned} \tag{4}$$

As shown in Table S5, the AIC favoured the three-way interaction model, whereas the BIC favoured the more parsimonious two-way interaction model. Given the additional complexity introduced by the three-way interaction, and the stronger penalty for complexity under BIC, we selected the simpler two-way interaction model.

Outcome-Feedback Model Selection

The outcome-feedback model also inherited the structure of the additive-feedback model but instead allowed the effect of lagged outcome to vary as a function of risk and communication, permitting an assessment of whether responsiveness to prior outbreak occurrence varies by treatment. As per the social-feedback model, to determine the appropriate model specification, we fit two models. The first model included the two-way interactions of lagged outcome with risk and communication. Formally, the two-way interaction model specification was:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{round} \\ & + \text{risk} : \text{communication} + \text{risk} : \text{round} + \text{communication} : \text{round} \\ & + \text{risk} : \text{communication} : \text{round} \\ & + \text{lagged_protection} + \text{lagged_outcome} \\ & + \text{risk} : \text{lagged_outcome} \\ & + \text{communication} : \text{lagged_outcome} \\ & + (1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player}) \end{aligned} \tag{5}$$

The second model included the same terms as the two-way interaction model, with the addition of the three-way interaction between risk, communication, and lagged outcome. Formally, the three-way interaction model specification was:

$$\begin{aligned} \text{decision} \sim\ & \text{risk} + \text{communication} + \text{round} \\ & + \text{risk} : \text{communication} + \text{risk} : \text{round} + \text{communication} : \text{round} \\ & + \text{risk} : \text{communication} : \text{round} \\ & + \text{lagged_protection} + \text{lagged_outcome} \\ & + \text{risk} : \text{lagged_outcome} \\ & + \text{communication} : \text{lagged_outcome} \\ & + \text{risk} : \text{communication} : \text{lagged_outcome} \\ & + (1 + \text{round} \mid \text{group}) + (1 + \text{risk} + \text{round} \mid \text{player}) \end{aligned} \tag{6}$$

As shown in Table S6, both the AIC and BIC unambiguously favoured the two-way interaction model and it was this model that we chose.

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Supplementary Tables

Table S1

Generic payoffs for the biosecurity game.

	COLUMN	
ROW	Biosecure	Not Biosecure
Biosecure	(BB, BN)	(BN, NB)
Not Biosecure	(NB, BN)	(NN, NN)

Table S2

Loss-averse type coordination game expected utility.

	COLUMN	
ROW	Biosecure	Not Biosecure
Biosecure	(10, 10)	(-5.15, 9.95)
Not Biosecure	(9.95, -5.15)	(2.92, 2.92)

Table S3

AIC and BIC for five different implementations of the treatment-only model predicting biosecurity decisions as a function of risk, communication, and round using different random effects structures at the player level. All models also included all two-way and three-way interactions between risk, communication, and round, with random intercepts for group and player. Bold values highlight the most successful model in terms of lowest AIC or BIC.

Model random effects	k	AIC	BIC
(1 | group) + (1 | player)	10	2695.56	2755.22
(1 | group) + (1 + risk | player)	12	2667.47	2739.06
(1 | group) + (1 + round | player)	12	2613.35	2684.94
(1 | group) + (1 + risk + round | player)	15	**2572.06**	**2661.54**
(1 | group) + (1 + risk : round | player)	19	2575.48	2688.83

Table S4

AIC and BIC for five different implementations of the treatment-only model predicting biosecurity decisions as a function of risk, communication, and round using different random effects structures at the group level. All models also included all two-way and three-way interactions between risk, communication, and round, with random intercepts for group and player, and by-player random slopes for risk and round. Bold values highlight the most successful model in terms of lowest AIC or BIC.

Model random effects	k	AIC	BIC
(1 | group) + (1 + risk + round | player)	15	2572.06	2661.54
(1 + risk | group) + (1 + risk + round | player)	17	2561.78	2663.20
(1 + round | group) + (1 + risk + round | player)	17	2546.35	**2647.77**
(1 + risk + round | group) + (1 + risk + round | player)	20	2540.00	2659.32
(1 + risk : round | group) + (1 + risk + round | player)	24	**2532.56**	2675.73

Table S5

AIC and BIC for two different versions of the social-feedback model predicting biosecurity decisions as a function of risk, communication, round, and the two-way and three-way interactions between these predictors. Both models also included the main effects of lagged outcome and lagged protection and the two-way interactions of lagged protection with risk and communication. The three-way interaction model additionally included the three-way interaction between risk, communication, and lagged protection. Both models included random intercepts for group and player, a by-group random slope for round, and by-player random slopes for risk and round. Bold values highlight the most successful model in terms of lowest AIC or BIC.

Model	k	AIC	BIC
Two-way interaction	21	2272.35	**2396.18**
Three-way interaction	22	**2270.56**	2400.29

Table S6

AIC and BIC for two different versions of the outcome-feedback model predicting biosecurity decisions as a function of risk, communication, round, and the two-way and three-way interactions between these predictors. Both models also included the main effects of lagged protection and lagged outcome and the two-way interactions of lagged outcome with risk and communication. The three-way interaction model additionally included the three-way interaction between risk, communication, and lagged outcome. Both models included random intercepts for group and player, a by-group random slope for round, and by-player random slopes for risk and round. Bold values highlight the most successful model in terms of lowest AIC or BIC.

Model	k	AIC	BIC
Two-way interaction	21	**2288.57**	**2412.39**
Three-way interaction	22	2289.76	2419.48

Table S7

Estimated generalized linear mixed-effects model parameters for the self-report model of risk and loss aversion predicting biosecurity decisions.

Fixed Effects					
Effect	β	SE	95% CI	z	p
Intercept	−2.11	0.39	[−2.88, −1.34]	−5.37	< .001
Risk	1.63	0.23	[1.18, 2.08]	7.03	< .001
Communication	1.84	0.54	[0.77, 2.90]	3.38	< .001
Risk Aversion	0.61	0.22	[0.18, 1.05]	2.75	.006
Loss Aversion	−0.05	0.20	[−0.45, 0.35]	−0.24	.812
Risk × Communication	0.62	0.32	[0.00, 1.24]	1.96	.050
Risk × Risk Aversion	0.03	0.16	[−0.29, 0.35]	0.20	.840
Communication × Risk Aversion	−0.60	0.26	[−1.10, −0.10]	−2.34	.019
Risk × Loss Aversion	0.13	0.16	[−0.18, 0.43]	0.81	.417
Communication × Loss Aversion	−0.07	0.25	[−0.56, 0.42]	−0.27	.790

Random Effects			
Level	Effect	Variance	SD
Player	Intercept	0.94	0.97
	Risk	1.05	1.02
Group	Intercept	1.35	1.16

Model Fit		
R^2	Marginal	Conditional
	0.29	0.59

Note—p-values for fixed effects were calculated using asymptotic Wald tests. Confidence intervals were calculated using the Wald method. The model included a by-player random slope for risk and random intercepts for player and group. The random-effect correlation was $r = -0.49$ for the player intercept–risk association. Risk aversion and loss aversion were standardised prior to analysis.

Table S8

Estimated generalized linear mixed-effects model parameters for the parameter-based model of risk and loss aversion predicting biosecurity decisions.

Fixed Effects					
Effect	β	SE	95% CI	z	p
Intercept	−2.07	0.41	[−2.87, −1.28]	−5.09	< .001
Risk	1.64	0.23	[1.20, 2.09]	7.19	< .001
Communication	1.83	0.56	[0.72, 2.94]	3.24	.001
Risk Aversion	0.64	0.20	[0.25, 1.03]	3.19	.001
Loss Aversion	0.15	0.19	[−0.24, 0.53]	0.75	.454
Risk × Communication	0.58	0.32	[−0.04, 1.21]	1.83	.067
Risk × Risk Aversion	−0.22	0.17	[−0.54, 0.11]	−1.31	.192
Communication × Risk Aversion	−0.55	0.25	[−1.05, −0.06]	−2.19	.029
Risk × Loss Aversion	−0.12	0.17	[−0.45, 0.21]	−0.69	.492
Communication × Loss Aversion	−0.06	0.25	[−0.55, 0.42]	−0.26	.799

Random Effects			
Level	Effect	Variance	SD
Player	Intercept	0.94	0.97
	Risk	1.00	1.00
Group	Intercept	1.48	1.22

Model Fit		
R^2	Marginal	Conditional
	0.28	0.59

Note—p-values for fixed effects were calculated using asymptotic Wald tests. Confidence intervals were calculated using the Wald method. The model included a by-player random slope for risk and random intercepts for player and group. The random-effect correlation was $r = -0.48$ for the player intercept–risk association. Risk aversion and loss aversion were standardised prior to analysis.

Supplementary Figures

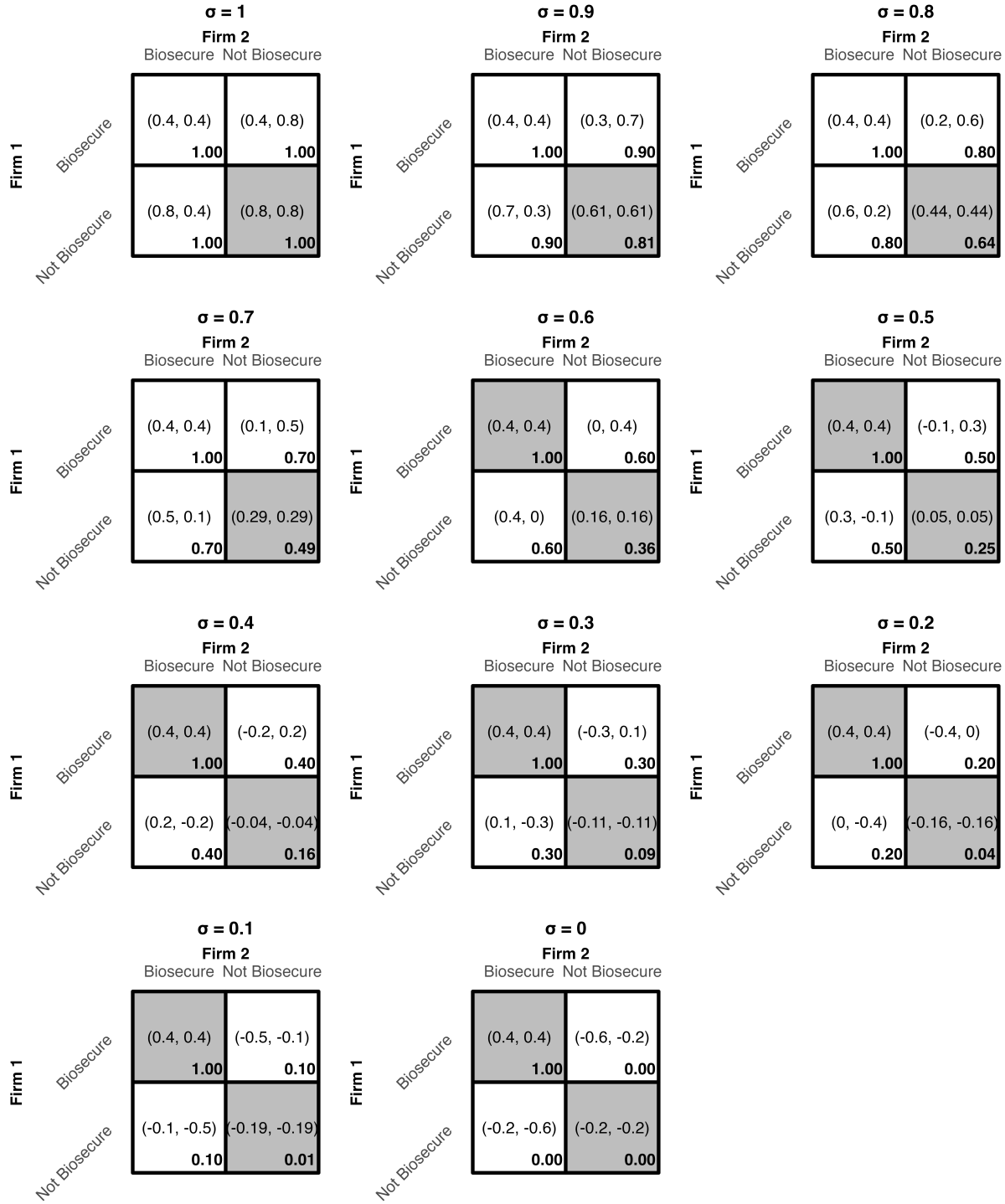


Figure S1

Expected payoffs and protection probabilities for different settings of the entry-threat parameter. Bold-face values represent protection probabilities, whereas shaded cells represent pure strategy Nash equilibria.